

Phase 3 — Summary of Attempts

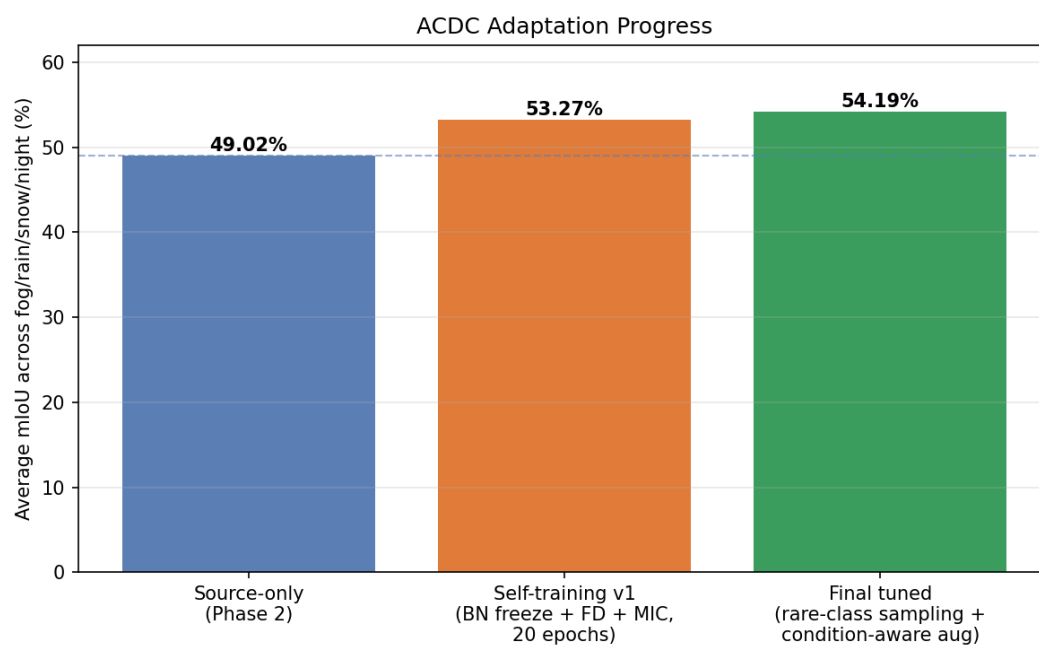
Self-training unsupervised domain adaptation, Cityscapes → ACDC

Eight training runs, spanning the attempts below, were made to close the domain gap identified in Phase 2 (49.02% average mIoU, zero-shot, across fog/rain/snow/night). Runs 2 and 3 diagnosed the same issue and are combined into one row. The final row is the best result obtained.

#	Attempt	What changed	Result	Outcome
1	Direct LR transfer	Phase 1's LR (6e-5) reused as-is	Collapsed by epoch 5	Failed — self-training collapse
2–3	Lower LR, night excluded	LR → 1.5e-5; night dropped from pseudo-label pool	Still degraded on all conditions	Failed — root cause not yet found
4	+ BatchNorm freeze	Froze BN layers on Phase 1's calibrated stats	+1.1 pt, stable training	Fix validated
5	+ feature-distance + MIC (first combined run)	20 epochs; night re-included	53.27% avg fog 67.4 / rain 55.2 / snow 55.9 / night 34.6	Best result at this point, but night still below its own 35.35% zero-shot baseline
6	Ablation study	3×6-epoch runs isolating each component	48.09% → 49.71% → 51.50%	Confirms combined config is strongest; plans a targeted night fix
7	First night-tuning attempt	+ rare-class sampling, consistency loss, night-aug, 14 epochs	Crashed (NameError), then unstable loss + cross-condition aug bug	Failed — 2 bugs found and fixed
8	Final tuned run (bugs fixed)	Same config as #7, corrected, 20 epochs	54.19% avg fog 67.4 / rain 56.6 / snow 57.8 / night 34.9	Best attempt overall

Best attempt: the final tuned run (#8)

54.19% average mIoU across fog/rain/snow/night — +5.17 points over the Phase 2 source-only baseline and +0.92 over the first combined run (#5). It is also the only configuration that improves fog, rain and snow simultaneously without sacrificing any of them: fog 67.4%, rain 56.6%, snow 57.8%, night 34.9%. Night remains marginally below its zero-shot baseline (35.35%) and is the one open item for further work.



Average mIoU at each major stage: source-only → first combined run (#5) → final tuned run (#8, best).